习题解答

☆ 8-5 (1) (–) Fe | $Fe^{2+}(c^{\ominus})$ || $Cu^{2+}(c^{\ominus})$ | Cu (+)

(2) (–) Zn | $Zn^{2+}(c^{\ominus})$ || $H^{+}(c^{\ominus})$ |$H_2(p^{\ominus})$ | Pt (+)

(3) (–) Cu | $Cu^{2+}(c^{\ominus})$ || $Fe^{2+}(c^{\ominus})$, $Fe^{3+}(c^{\ominus})$ | Pt (+)

8-6 (1) $E^{\ominus} = \varphi^{\ominus}_{+} - \varphi^{\ominus}_{-}$

$= \varphi^{\ominus}(Fe^{2+}/Fe) - \varphi^{\ominus}(Mg^{2+}/Mg) = 1.925\ V$

(2) $E^{\ominus} = \varphi^{\ominus}_{+} - \varphi^{\ominus}_{-}$

$= \varphi^{\ominus}(Ag^{+}/Ag) - \varphi^{\ominus}(AgI/Ag) = 0.9518\ V$

8-7 (1) $E^{\ominus} = \varphi^{\ominus}(Zn^{2+}/Zn) - \varphi^{\ominus}(Cd^{2+}/Cd) < 0$ 逆向！

(2) $E^{\ominus} = \varphi^{\ominus}(Ag^{2+}/Ag) - \varphi^{\ominus}(Sn^{4+}/Sn^{2+}) > 0$ 正向！

(3) $E^{\ominus} = \varphi^{\ominus}(IO_3^{-}/I^{-}) - \varphi^{\ominus}(Br_2/Br^{-}) > 0$ 正向！

(4) $E^{\ominus} = \varphi^{\ominus}(MnO_4^{-}/MnO_2) - \varphi^{\ominus}(MnO_2/Mn^{2+}) > 0$ 正向！

习题解答

☆ 8-10 (1) $I_2 + Cd = Cd^{2+} + 2I^-$

+: $I_2 + 2e = 2I^-$ –: $Cd = Cd^{2+} + 2e$

(–) $Cd \mid Cd^{2+}(c^\ominus) \parallel I^-(c^\ominus) \mid I_2(s) \mid Pt$ (+)

(2) $2Fe^{3+} + H_2 = 2Fe^{2+} + 2H^+$

+: $Fe^{3+} + e = Fe^{2+}$ –: $H_2 = 2H^+ + 2e$

(–) $Pt \mid H_2(p^\ominus) \mid H^+(c^\ominus) \parallel Fe^{3+}(c^\ominus), Fe^{2+}(c^\ominus) \mid Pt$ (+)

(3) $\varphi(Fe^{3+}/Fe^{2+}) = \varphi^\ominus(Fe^{3+}/Fe^{2+}) - \frac{RT}{F}\ln\frac{c(Fe^{2+})}{c(Fe^{3+})} = 0.6526$

$2Fe^{3+} + 2I^- = 2Fe^{2+} + I_2$

+: $Fe^{3+} + e = Fe^{2+}$ –: $2I^- = I_2 + 2e$

(–) $Pt \mid I_2(s) \mid I^-(c^\ominus) \parallel Fe^{3+}(c^\ominus), Fe^{2+}(c^\ominus) \mid Pt$ (+)

☆ 8-13 解： (1) (–) Zn | Zn^{2+}(2.0M) || Fe^{2+} (2.0×10^{-12}M) |Fe (+)

+: $Fe^{2+} + 2e = Fe$ –: $Zn = Zn^{2+} + 2e$

$$Fe^{2+} + Zn = Zn^{2+} + Fe$$

(2) $$E = E^{\ominus} - \frac{RT}{2F}\ln\frac{c(Zn^{2+})}{c(Fe^{2+})} = \varphi_+^{\ominus} - \varphi_-^{\ominus} - \frac{RT}{2F}\ln\frac{c(Zn^{2+})}{c(Fe^{2+})}$$

$$= -0.447 - (-0.762) - \frac{0.0592}{2}\lg\frac{2.0}{2.0\times10^{-12}} = 0.670\text{V}$$

(3) $E = 0.670\ \text{V} > 0$ 反应正向自发！

原电池可以向外释放电能！

习题解答

☆ 8-16 解： (1) (–) Cu | Cu^{2+}(0.5M) || Ag^{+}(0.5M) |Ag (+)

(2) +: $Ag^{+} + e = Ag$ –: $Cu = Cu^{2+} + 2e$

$$2Ag^{+} + Cu = Cu^{2+} + 2Ag$$

(3) $$E = E^{\ominus} - \frac{RT}{2F}\ln\frac{c(Cu^{2+})}{c^{2}(Ag^{+})} = \varphi_{+}^{\ominus} - \varphi_{-}^{\ominus} - \frac{RT}{2F}\ln\frac{c(Cu^{2+})}{c^{2}(Ag^{+})}$$

$$= 0.7996 - 0.3419 - \frac{0.0592}{2}\lg\frac{0.50}{0.50^{2}} = 0.4488V$$

(4) $$-nFE^{\ominus} = -RT\ln K^{\ominus}$$

$$2\times96500\times(0.7996 - 0.3419) = 8.314\times298.2\ln K^{\ominus}$$

$$K^{\ominus} = 2.9\times10^{15}$$

习题解答

☆ 8-18 解： (1) (–) Pt | $Fe^{3+}(c^{\ominus}), Fe^{2+}(c^{\ominus})$ || $Ag^{+}(c^{\ominus})$ |Ag (+)

(2) $E^{\ominus} = \varphi^{\ominus}(Ag^{+}/Ag) - \varphi^{\ominus}(Fe^{3+}/Fe^{2+}) = 0.7996 - 0.771 = 0.0286\ V$

$\Delta_r G_m^{\ominus} = -nFE^{\ominus} = -1\times96500\times0.0286 = -2760\ J/mol$

$-nFE^{\ominus} = -RT\ln K^{\ominus}$

$1\times96500\times0.0286 = 8.314\times298.2\ln K^{\ominus}$

$K^{\ominus} = 3.04$

(3)

	Ag^{+}	+	Fe^{2+} =	Ag	+	Fe^{3+}
c_{eq}	1.0		0.1–x			x

$K^{\ominus} = 3.04 = x/\{1.0\times(0.1-x)\}$ $x = 0.75$ $\alpha = 75\%$

(4) $E = E^{\ominus} - 0.0592\lg\dfrac{c(Fe^{3+})}{c(Fe^{2+})c(Ag^{+})} = \varphi_{+}^{\ominus} - \varphi_{-}^{\ominus} - 0.0592\lg\dfrac{1.0}{1.0\times c(Ag^{+})}$

$= 0.0286 - 0.0592\lg\dfrac{1.0}{1.0\times\dfrac{c_{eq}([Ag(NH_3)_2]^{+})}{K_f^{\ominus}\times c_{eq}^{2}(NH_3)}} == 0.0286 - 0.0592\lg(K_f^{\ominus}\times0.10^{2})$

< 0 不自发！

习题解答

8-21 解： 电极 $2H^+ + 2e = H_2$

$$HAc + H_2O = Ac^- + H_3O^+$$

$$\varphi(H^+/H_2) = \varphi^{\ominus}(H^+/H_2) - \frac{0.0592}{2}\lg\frac{p(H_2)/p^{\ominus}}{c^2(H^+)}$$

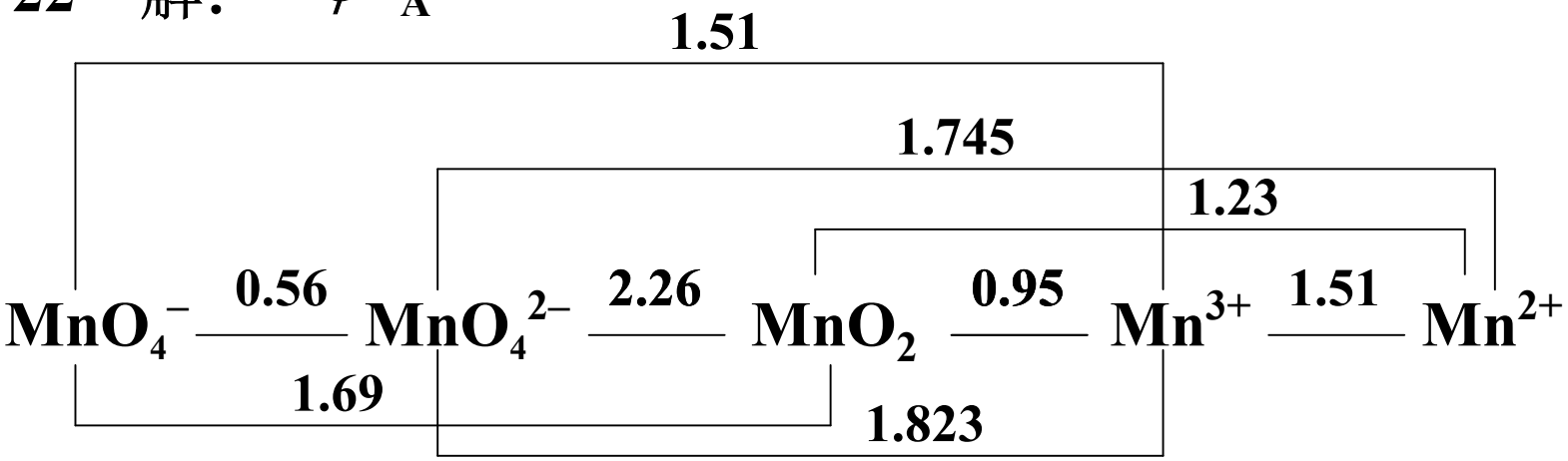
$$= 0 - 0.0592\lg c(H^+)$$

$$= 0.0592\lg\sqrt{c_0(HAc)\times K_a^{\ominus}(HAc)}$$

$$= 0.0592\lg\sqrt{0.10\times 1.8\times 10^{-5}}$$

$$= -0.17V$$

8-22　解：　$\varphi^{\ominus}_{A}$



(1)　$\varphi^{\ominus}(MnO_4^-/Mn^{2+}) = (0.56\times1+2.26\times2+0.95\times1+1.51\times1)/5$

$= 1.508\ V$

(2) MnO_2不能歧化

(3) 能歧化的是：　MnO_4^{2-}和Mn^{3+}反应如下：

MnO_4^{2-}:

$3MnO_4^{2-} + 4H^+ = 2MnO_4^- + MnO_2 + 2H_2O$

$4MnO_4^{2-} + 8H^+ = 3MnO_4^- + Mn^{3+} + 4H_2O$

$5MnO_4^{2-} + 8H^+ = 4MnO_4^- + Mn^{2+} + 4H_2O$

Mn^{3+}:

$Mn^{3+} + 2H_2O = MnO_2 + Mn^{2+} + 4H^+$

$5Mn^{3+} + 4H_2O = MnO_4^- + 4Mn^{2+} + 8H^+$ (有可能)